



PAPER – 3: QUANTITATIVE APTITUDE



QUESTIONS

- 4 tables and 3 chairs together cost ₹ 2,250 and 3 tables and 4 chairs cost ₹ 1950. Find the cost of 2 chairs and 1 table.
(A) ₹ 550
(B) ₹ 1050
(C) ₹ 750
(D) None of these
- One root of the equation: $x^2 - 2(5 + m) + 3(7 + m) = 0$ is reciprocal of the other. Find the value of m.
(A) $-20/3$
(B) 7
(C) $1/7$
(D) 117
- Sangeeta leaves from her home. She first walk 30 metres in North-West direction, and then 30m in South-West direction, next she walks 30 metres in South-East direction. Finally she turns towards her house. In which direction is she moving?
(A) North-West
(B) North-East
(C) South-East

- (D) South-West
4. If a simple interest on a sum of money at 6% p.a. for 7 years is equal to twice of simple interest on another sum for 9 years at 5% p.a. The ratio will be:
- (A) 2:15
(B) 7:15
(C) 15:7
(D) 1:7
5. The Scarap value of machine valued at Rs,10,00,000 after 15 years of depreciation is 10% per annum.
- (A) ₹ 215891.13
(B) ₹ 205891.13
(C) ₹ 225891.13
(D) None
6. If the difference between the compound interest compounded annually and simple interest on a certain amount at 10% per annum for two years is ₹ 372, then the principal amount is.
- (A) ₹ 37,000
(B) ₹ 37,200
(C) ₹ 37,500
(D) None of the above
7. Find the present value of an annuity of ₹ 1,000 payable at the end of each year for 10 years. If rate of interest is 6% compounding per annum. (given $(1.06)^{-10} = 0.5584$):
- (A) ₹ 7,360
(B) ₹ 8,360

- (C) ₹ 12,000
- (D) None of these.
8. What will be the population after 3 years, when present population is 1,00,000 and the population increases at 3% in year 1st year, at 4% in second year and 5% in third year.
- (A) 1,12,476
- (B) 1,15,476
- (C) 1,20,576
- (D) 1,25,600
9. Find the present value of an annuity which pays 200 at the end of each 3 months for 10 years assuming money to be worth 5% converted quarterly?
- (A) ₹ 3473.86
- (B) ₹ 3108.60
- (C) ₹ 6265.38
- (D) None of these
10. The value of furniture depreciates by 10% a year, if the present value of the furniture in an office is ₹ 21870, calculate the value of furniture 3 years ago:
- (A) ₹ 30,000
- (B) ₹ 40,000
- (C) ₹ 35,000
- (D) ₹ 50,000
11. The letters of the word VIOLENT are arranged so that the vowels occupy even place only. The number of permutations is:
- (A) 144

- (B) 120
(C) 24
(D) 72
12. If the p th term of an A.P. is 'q' and the q th term is 'p', then its r th term is:
(A) $p + q + r$
(B) $p + q - r$
(C) $p - q - r$
(D) $p + q$
13. Find the sum of all natural numbers between 250 and 1,000 which are exactly divisible by 3:
(A) 1,56,375
(B) 1,56,357
(C) 1,65,375
(D) 1,65,357
14. Let R is the set of real numbers, such that the function $f: R \rightarrow R$ and $g: R \rightarrow R$ are defined by $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$. Find (fog) :
(A) $4x^2 + 6x + 1$
(B) $x^2 + 6x + 1$
(C) $4x^2 - 6x + 1$
(D) $x^2 - 6x + 1$
15. Find the positive value of k for which the equations: $x^2 + kx + 64 = 0$ and $x^2 - 8x + k = 0$ will have real roots:
(A) 12
(B) 16
(C) 18

- (D) 22
16. Praveen is facing west. He turns 45° in the clockwise direction and then again another turns with 180° in the same direction i.e. clockwise direction, after that he turns 270° in the anti-clockwise direction. Which direction is he facing now?
- (A) North-West
(B) West
(C) South-West
(D) South
17. If $\log_4(x^2 + x) - \log_4(x + 1) = 2$ then the value of x is
- (A) 2
(B) 3
(C) 16
(D) 8
18. If HEALTH is written as GSKZDG, then how will NORTH be written in that code?
- (A) OPSUI
(B) GSQNM
(C) FRPML
(D) IUSPO
19. In a certain code, TEACHER is written as VGCEJGT. How is CHILDREN written in that code?
- (A) EJKNEGTP
(B) EGKNEITP
(C) EJKNFGTO
(D) EJKNFTGP

20. The covariance between two variables X and Y is 8.4 and their variances are 25 and 36 respectively. Calculate Karl Pearson's coefficient of correlation between them.
- (A) 0.82
(B) 0.28
(C) 0.01
(D) 0.09
21. If two variables x and y are related by $2X + 3Y - 7 = 0$ and the mean and mean deviation about mean of X are 1 and 0.3 respectively, then the coefficient of mean deviation of Y about mean is.
- (A) -5
(B) 4
(C) 12
(D) 50
22. The mean salary for a group of 40 female workers is ₹ 5200 per month and that for a group of 60 male workers is ₹ 6800 per month. What is the combined salary?
- (A) ₹ 6160
(B) ₹ 6280
(C) ₹ 6890
(D) ₹ 6920
23. The average age of 15 students is 15 years. Out of these the average age of 5 students is 14 years and that of other 9 students is 16 years, then the age of 15th student is ____
- (A) 11 years
(B) 14 years
(C) 15 years

- (D) None of these
24. The standard deviation of 25, 32, 43, 53, 62, 59, 48, 31, 24, 33 is
- (A) 13.23
- (B) 12.33
- (C) 11.33
- (D) none of these
25. If the range of x is 2, what would be the range of $-3x + 50$?
- (A) 2
- (B) 6
- (C) -6
- (D) 44
26. The Standard deviation is independent of change of
- (A) Origin
- (B) Scale
- (C) Both
- (D) none
27. In _____ distribution, mean = variance.
- (A) Normal
- (B) Binomial
- (C) Poisson
- (D) none of these
28. The pair of averages whose value can be determined graphically.
- (A) Mean and Median
- (B) Mode and Mean

- (C) Mode and Median
 - (D) None of these
29. The tests of shifting bases are called _____
- (A) Unit test
 - (B) Time reversal test
 - (C) Circular test
 - (D) None
30. Purchasing power of money is stated as _____ price index?
- (A) Equal to
 - (B) Reciprocal of
 - (C) Unequal to
 - (D) None